LALIGENCE

LALIGENCE eKYC Project Overview

DOCUMENT PROOFING | LIVENESS | FACE ID | RISK DECISION

A web-first identity verification and face-login prototype for passport/Lao ID onboarding, live user proofing, selfie face matching, fraud risk decisions, and returning-user Face ID authentication.

End-to-end workflow

Start → Document → Liveness → Gesture → Selfie → Decision → Face ID

User Journey

- **Start / New user**
  Create a verification session with a generated or supplied user ID.
- **Document proofing**
  Upload or camera-capture passport/Lao ID evidence.
- **Active liveness**
  Perform random blink, mouth, and head-movement checks.
- **Gesture challenge**
  Show 3 random hand gestures inside a moving target circle.
- **Selfie match**
  Capture a live selfie and compare it to the document portrait.
- **Result**
  Return pass/reject, reason codes, OCR, risk, and match scores.
- **Face ID**
  Enroll a verified face for login and Face Pay-style approval.

Security Intelligence

- **OCR/MRZ**
  Reads passport MRZ and key ID fields; validates expiry and check digits when available.
- **Document fraud**
  Scores quality, document-likeness, tamper cues, recapture risk, portrait substitution, and printed-copy risk.
- **Face recognition**
  Uses OpenCV YuNet + SFace for document/selfie embeddings and comparison.
- **Passive liveness**
  Combines ONNX anti-spoof models with screen, phone, paper, glare, and moire heuristics.
- **Step control**
  Locks later steps until previous proofing steps pass.
- **Auto rejection**
  High-risk signals produce explainable reason codes instead of silent failure.

Platform & Testing

- **Backend**
  Python FastAPI API with local model adapters and configurable thresholds.
- **Frontend**
  React/Vite camera workflow with branded UX and mobile-aware capture.
- **Storage**
  SQLite locally and PostgreSQL in deployment through SQLAlchemy.
- **SDKs**
  Flutter and TypeScript SDKs wrap the same backend API.
- **Datasets**
  Scripts support document fraud, print-copy, selfie spoof, and camera collection tests.
- **Deployment**
  Render, DigitalOcean/Caddy/PostgreSQL, Vercel, and Cloudflare tunnel options.

Data and security note

Raw uploads are analyzed in-memory during normal verification. Enrolled profiles store verified fields and face templates. Production hardening still needs encrypted biometric storage, consent/retention policy, audit logging, protected admin APIs, and larger calibrated validation datasets.